

MECH 410

Analysis and Optimization of a Brake Rotor for FSAE Applications

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Abstract

This project presents a thermal and structural analysis of an FSAE brake rotor using Siemens NX. Following recorded operating temperatures exceeding 600°C during competition, transient thermal and structural FEA simulations were conducted to verify the thermal performance of the existing design. Two alternative rotor concepts, drilled and slotted, were developed and evaluated against defined design objectives of peak operating temperature, structural integrity, and minimum mass. Geometry optimization was performed on each concept and results were compared to identify the optimal rotor design for FSAE competition use.

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1.0 Introduction

1.1 General Background and Problem Definition

All members of this project group are a part of the Formula SAE (FSAE) team at the University of Victoria. An aspect of the car which our team has proposed to research has been the braking system, specifically the brake rotors. Under racing conditions, brake rotors are exposed to extreme and repeated forces where 100% of the energy is converted to heat. Due to this extreme heat, if brake rotors are not designed accordingly, failure due to thermal overload may occur causing extreme safety issues for the car in motion.

During competition at FSAE Michigan in 2025, the brake rotor temperature during the endurance event was measured to be ~630°C. This far exceeds the ideal operating temperature for the 4130 steel rotors (approximately 100-300°C) [1], increasing the risk of thermal cracking and permanent deformation. High rotor temperatures can also negatively affect nearby components such as bearings, calipers, and brake fluid, reducing their lifespan and performance. Due to this, it is important that the brake rotors are redesigned and optimized to perform to the demands of a racing environment.

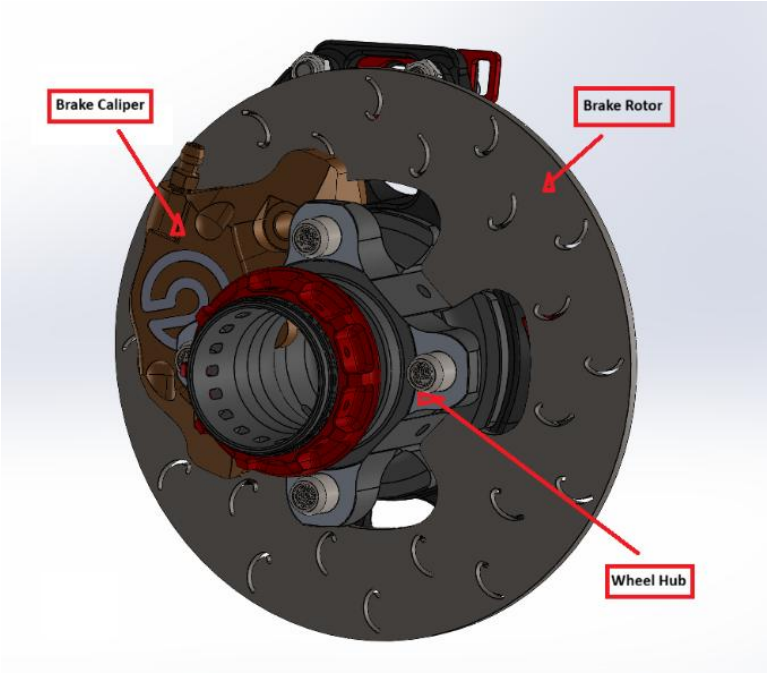


Figure 1: UV25 Corner Assembly

1.2 Project Objectives

The key objectives of this project surround the optimization of the brake rotor for its intended use case. Specifically, redesigning the brake rotors to be utilized on the front hubs and rear axle of the 2026 UVic FSAE car. The redesigned rotors will aim to achieve at least a 10-15% mass reduction compared to the current design. Thermal performance will be improved by decreasing the

maximum steady-state operating temperature to remain under 500°C under repeated braking conditions. The design must also be fully compatible with the existing mounting geometry without requiring modification to mounting components.

2.0 Project Scope

The scope of this project involves all aspects of creation and optimization of the brake rotors for an FSAE vehicle. This begins with material selection for thermal properties and coefficient of friction. Next, modelling and FEA simulations will be run to determine the maximum stress within the part under the specified loading. Thermal simulations will be conducted under transient conditions to determine the thermal performance and balance from the input heat flux and convective cooling. Lastly, the overall dimensions of the part (outer diameter and thickness), as well as the internal lightweighting geometries, will be varied based on the FEA and thermal results to fully optimize the part for prescribed design objectives.

3.0 CAD/CAE Modelling Tool Selection

All previous part models and assemblies for the UVic FSAE team have been developed in SolidWorks 2022 PDM. For this project, the brake rotor will be initially modeled and analyzed in NX. After completion of the project, it will be recreated in SolidWorks to ensure compatibility with the UV26 car.

NX was selected for the modeling and simulation work to expand the group's experience with advanced CAD/CAE tools beyond SolidWorks. This allows the team to build on knowledge developed in MECH 410 while gaining experience in a different software environment.

4.0 Technical Challenges and Special Features

4.1 Preliminary Calculations

A large technical challenge of this project was determining the magnitude of the required loads for the brake rotor. Parameters needed include magnitudes of brake pad pressure and friction force, heat flux, and convection coefficients. To begin, the following UV25 parameters will be utilized.

Table 1: UV25 Parameters - Structural Calculations

UV25 Parameter	Value
Brake Piston Diameter [m]	0.034
Brake Pad Area [m^2]	0.0016
Brake Piston Area [m^2]	9.08E-04
Brake Pressure [Pa]	4.14E+06
Brake Pad Coefficient of Friction	0.5

Using the values listed above, the total force applied by the brake pads can be calculated. It is important to note that the brake caliper consists of two opposing pistons, therefore the area is doubled.

$$F_{pad} = \frac{P_{brake}}{2A_{piston}} = \frac{4.14 * 10^6\ Pa}{2 * (9.08 * 10^{-4} m^2)} = 7517.6\ N$$

As these values will be utilized in the FEA analysis, the force per brake pad can be calculated.

$$F_{pad,1} = F_{pad,2} = \frac{1}{2}F_{pad} = \frac{1}{2}(7517.6\ N) = 3758.8\ N$$

Using the force per brake pad, the applied pressure per brake pad can be determined.

$$P_{pad,1} = P_{pad,2} = \frac{F_{pad,1}}{A_{pad}} = \frac{F_{pad,2}}{A_{pad}} = \frac{7517.6\ N}{0.0016\ m^2} = 2.35\ MPa$$

Lastly, using the manufacturer specified brake pad coefficient of friction [1], the friction force applied by the brake pads can be calculated.

$$F_f = \mu F_{pad,1} = \mu F_{pad,2} = 0.5(3758.8\ N) = 1879.4\ N$$

With the mechanical loads established, the thermal conditions were determined using the parameters outlined in Table 2 below.

Table 2: UV25 Parameters - Thermal Calculations

UV25 Parameters	**Value**
Rotor Diameter [m]	0.1778
Rotor Mass [kg]	0.535
Tire Radius [m]	0.1143
Average Velocity [m/s]	17.25
Maximum Velocity [m/s]	21.39
Minimum Velocity [m/s]	12.36
Braking Time [s]	1.47
Mass Vehicle + Driver [kg]	269.80
Brake Bias [%]	50.0

Brake rotor research studies, specifically for FSAE applications, were reviewed and equations from these articles were used to determine the required loads. To begin, the Reynold's number and convection coefficient can be calculated using equations from [2].

$$Re = \frac{\frac{V_{avg}}{R_{tire}}{R_{disc}}^2}{\upsilon} = \frac{\frac{17.25\ \frac{m}{s}}{0.1143\ m}0.0889\ m^2}{1.93 * 10^{-5}\frac{kg}{m * s}} = 6.18 * 10^4$$

$$h_{conv} = 3.974\frac{k}{D}Re^{0.55} = 3.974\frac{0.026\frac{W}{m*K}}{0.1778\,m}(6.18*10^4)^{0.55} = 250.77\frac{W}{m^2K}$$

With the convection parameters determined, the heat flux applied to the brake rotor must be calculated. Using equations derived from [2] paired with basic heat transfer principles, the heat flux per rotor can be determined.

$$KE = \frac{m({V_{max}}^2 - {V_{min}}^2)}{2} = \frac{269.8\,kg((21.39\frac{m}{s})^2 - (12.36\frac{m}{s})^2)}{2} = 41098\,J$$

$$KE_{rotor} = \eta_{bias} * KE = 0.5 * 41098\,J = 10274.5\,J$$

$$P_{rotor} = \frac{KE}{\Delta T} = \frac{10274.5\,J}{1.47\,s} = 14002.7\,W$$

$${q_{rotor}}'' = \frac{P}{A} = \frac{14002.7\,W}{42668.4\,mm^2} = 0.328\,\frac{W}{mm^2}$$

With all thermal and structural loads determined, the simulation is ready to be prepared.

4.2 Material Selection

As 4130 steel (existing rotor material) was unavailable in the Siemens NX material library, 4340 steel was selected as a substitute for all simulations conducted. The key thermal and structural properties of both materials are compared in Table 3. The thermal properties which govern simulation accuracy (thermal conductivity, specific heat capacity, and the thermal expansion coefficient) are virtually identical between the two alloys, confirming thermally accurate material substitution. Both materials also share the same Young's modulus, ensuring valid structural stress representation. Notably, 4340 steel possesses a higher yield strength than 4130, meaning the structural analysis is non-conservative. However, given that the primary focus of this study is thermal performance rather than structural assessment, this difference has negligible impact on the validity of the simulation results and conclusions presented. As a result, 4340 steel is considered a valid substitute for use throughout this analysis.

Table 3: 4130 Steel vs. 4340 Steel Material Properties

	4130 Steel	**4340 Steel**
Young's Modulus [GPa]	200-210	205-213
Yield Strength [MPa]	483-534	770-950
Thermal Conductivity [W/m-K]	41.5	35-45
Specific Heat Capacity [J/kg-K]	461-480	440-520
Thermal Expansion Coefficient [$°C^{-1}$]	10^{-6}	10^{-6}

4.3 Thermal Analysis

As the main purpose of a brake rotor on a vehicle is to convert mechanical energy into heat energy, thermal loads and analysis will be integral to the operation of the brake rotor. A transient thermal

simulation (SOL 159) was conducted to replicate worst-case operating conditions experienced during the FSAE endurance event. Two thermal boundary conditions were applied: convective cooling across all exposed rotor surfaces, and heat flux applied to the fixed brake pad contact region. Convection represents the continuous heat dissipation to the surrounding airflow throughout the event, while heat flux represents heat generation during braking.

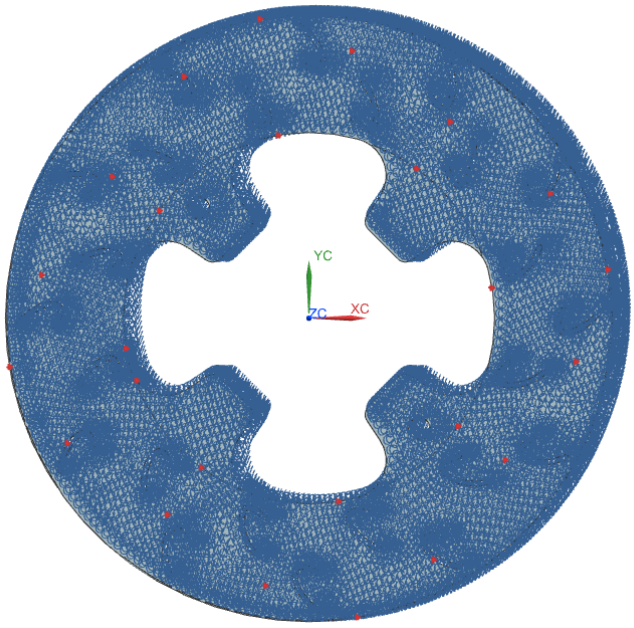


Figure 2: Thermal Loading and Boundary Conditions

Due to the cyclical nature of braking, heat flux was applied as a time-varying step function using a tabular field defined within the NX simulation environment. The heat flux profile was constructed to represent a single endurance lap of 140.7 seconds (taken from 2025 competition data), incorporating small (one second), medium (two seconds), and large (three seconds) braking events corresponding to the variety of corner types and straight lengths encountered on a typical FSAE endurance track. During non-braking periods the heat flux is zero, allowing the rotor to dissipate heat through convection alone. A portion of the applied heat flux profile is shown in Figure 3 below.

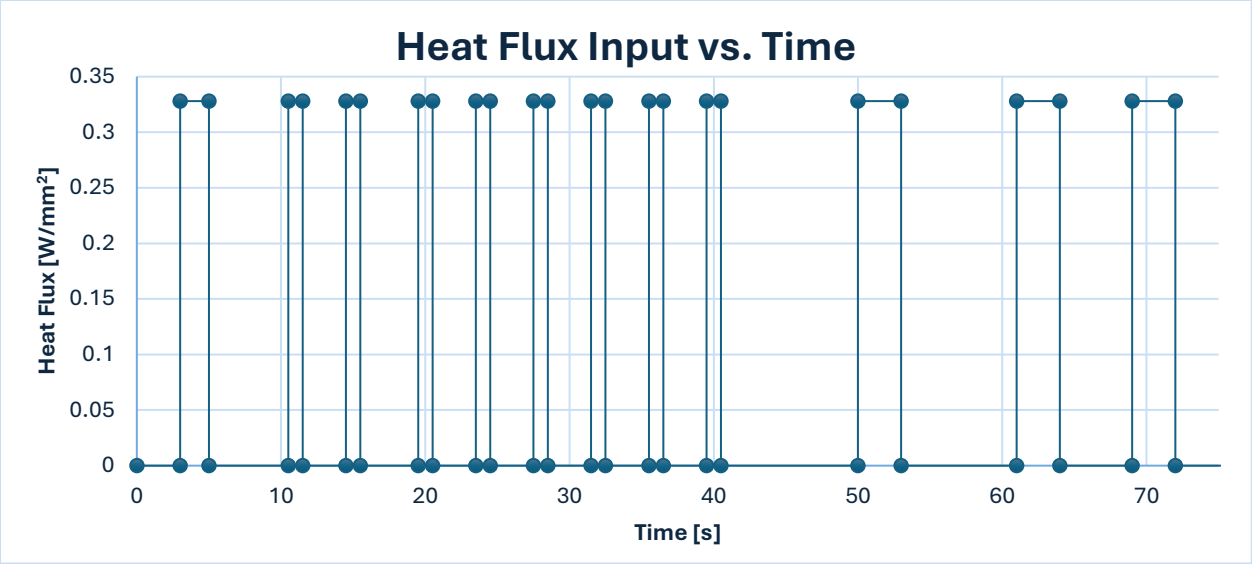


Figure 3: Heat Flux vs. Time Plot

The following thermal plot was chosen from one of the increments of the transient thermal analysis (Increment 2, time = 4.9s). While the magnitudes of the temperatures change throughout each increment, the temperature distribution remains consistent with the hottest area being the contact patch of the brake pads, and the coolest location being the mounting clovers. A graph of temperature versus time was created for the hottest node of the rotor.

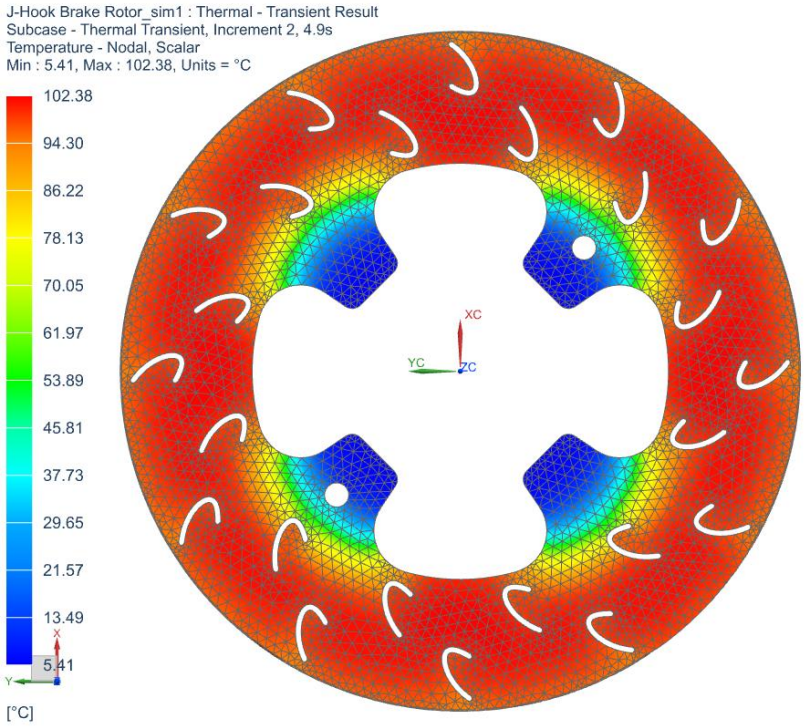


Figure 4: J-Hook Transient Thermal Results

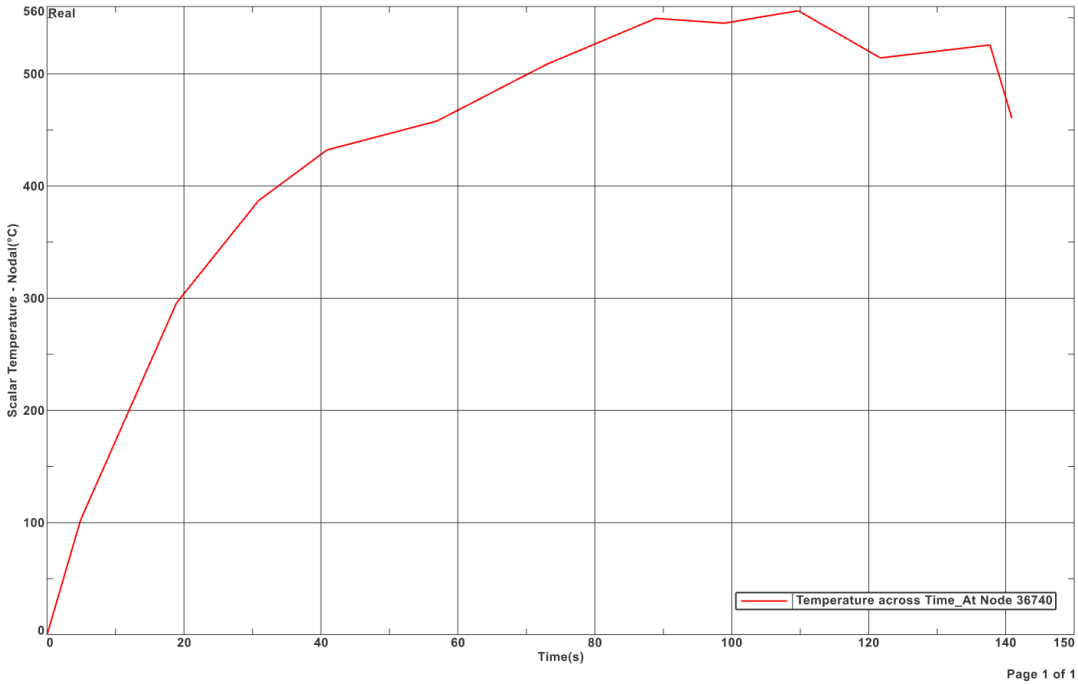


Figure 5: Temperature vs. Time Plot

From the transient thermal results, the peak operating temperature of the original J-hook rotor was approximately 560°C, occurring at the brake pad contact surface. It is important to note that during competition the brake rotors were recorded reaching 630°C across multiple laps (the 560°C result represents only a single lap) and temperatures would be expected to accumulate further as the rotor operates over the duration of the endurance event. This is consistent with the recorded competition data, validating the simulation methodology and boundary conditions applied. The resulting temperature distribution was imported into the structural FEA and optimization simulations as the thermal loading input.

4.4 Structural FEA

Structural simulations were performed on the J-hook rotor concept. As the rotor utilises a floating mount configuration, no bolts are required to secure the rotor to the wheel hub. Fixed constraints were therefore applied to the clover mounting faces that contact the wheel hub. While a fixed constraint does not perfectly represent the mounting condition, it is the most appropriate boundary condition available within the FEA environment and is standard practice for this type of analysis.

Three loads were applied to the structural simulation: brake pad contact pressure, frictional force, and thermal loading imported from the transient thermal analysis. The contact pressure was applied normal to the brake pad contact faces, and the frictional load was applied tangentially to the same surfaces. Both loads were calculated as outlined in the previous section. Thermal loads were imported directly from the previous thermal solution and applied to the entire rotor body, representing the temperature distribution at peak operating conditions.

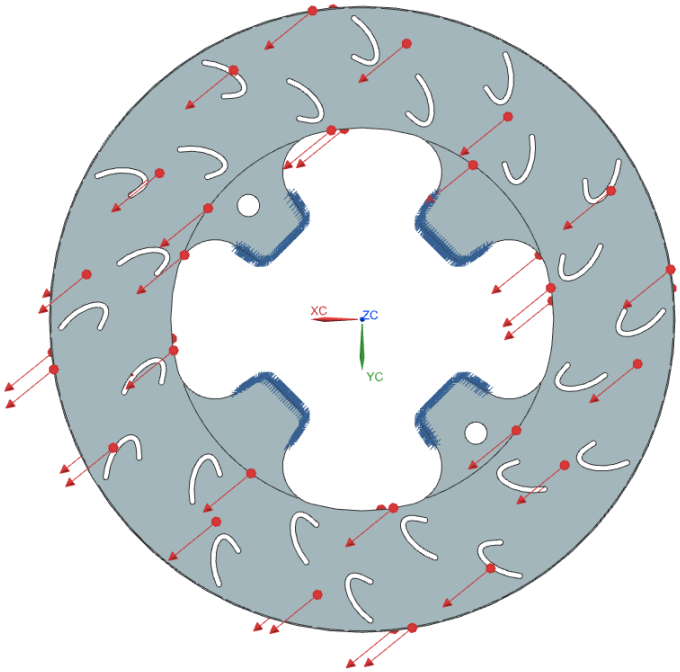


Figure 6: FEA Loading and Boundary Conditions

After defining the loads and boundary conditions, the FEA simulation was run. The stress distribution across the J-hook rotor is shown in the figure below. The majority of the rotor body experiences relatively low stress, with concentrations occurring at the small radii of the J-hook geometry and at localized regions of the mounting clover.

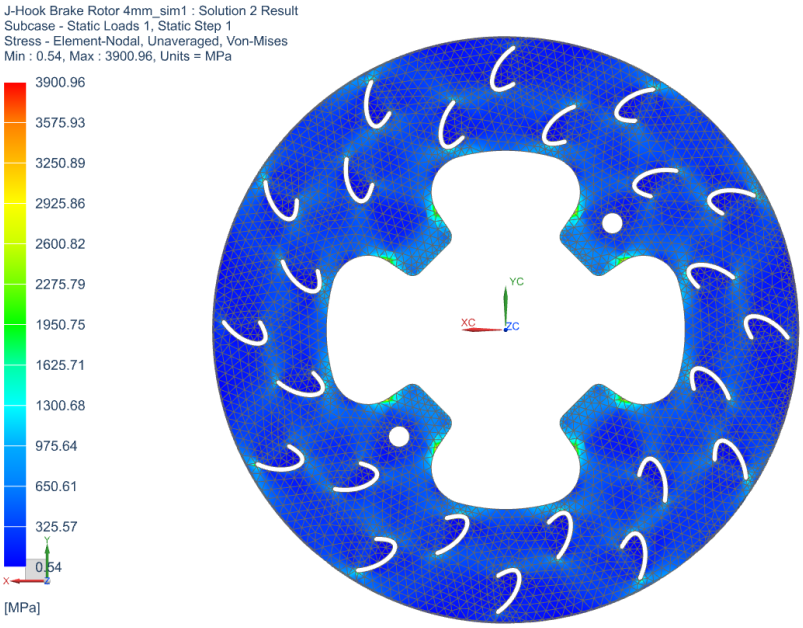


Figure 7: J-Hook Rotor Stress Distribution

Thermal loading was found to dominate the structural response, with the applied contact pressure and frictional loads contributing essentially negligible stress by comparison. This is consistent with the high peak operating temperatures experienced by the rotor during the simulated endurance event, where thermal expansion and temperature gradients across the rotor body generate significantly greater stress than the mechanical braking loads alone.

Brake rotors are designed to operate under conditions where localized yielding is expected and acceptable. Under peak thermal loading, elevated operating temperatures reduce the material yield strength, making a factor of safety greater than 1 unachievable across the entire rotor body. Brief localized yielding during extreme braking events is considered normal behaviour in brake rotor design [3]. The structural assessment therefore focuses on minimizing peak stress concentrations and achieving the most uniform stress distribution possible across the rotor, rather than targeting a static factor of safety above 1.

Figure 8 illustrates a stress singularity present in the FEA results, occurring at the location where the rotor geometry transitions from curved to flat located at the edge of the fixed constraint boundary condition. Mesh refinement was applied at this location in an attempt to resolve the singularity, however, the stress continued to increase with refinement, confirming the mathematical singularity rather than a true stress concentration. Stress singularities are a known limitation of FEA and do not represent a physically meaningful stress state. As the stress at these locations significantly exceed the yield strength of the rotor material, they are excluded from the structural analysis. All design evaluation and comparison between rotor concepts is based solely on the stress distribution observed throughout the rotor body.

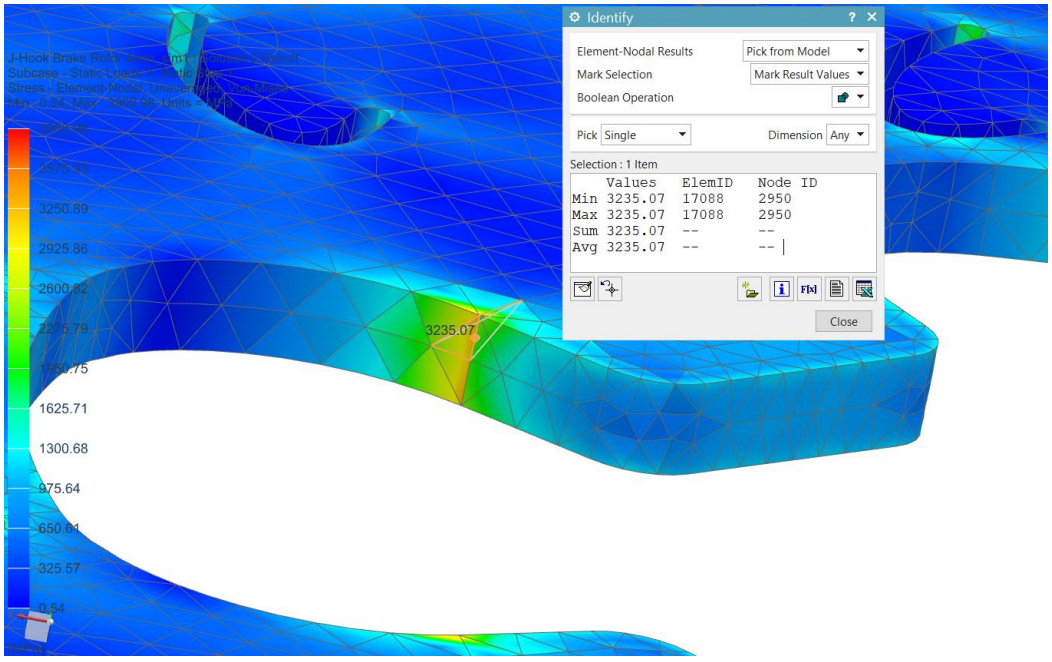


Figure 8: J-Hook Rotor Stress Singularity

The figure below illustrates a true stress concentration occurring at the radii of the J-hook geometry. This stress concentration is physically meaningful and represents a legitimate area of structural concern. The concentration arises due to the small radii inherent in the J-hook, where the geometry profile restricts the rotor's ability to expand under thermal loading. The elevated stress at this location raises concern for thermal cracking under repeated high temperature braking cycles, as the cyclical stress associated with each braking and cooling event could propagate cracks through the rotor body over repeated use.

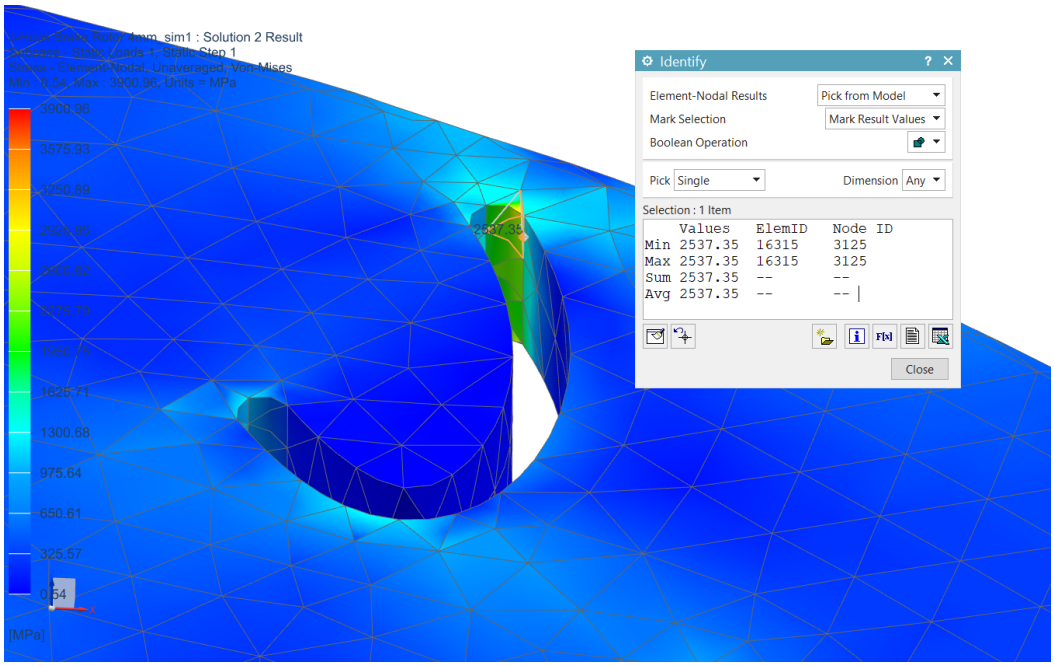


Figure 9: J-Hook Rotor Stress Concentration

4.5 Alternative Design Concepts

Having validated the thermal and structural limitations of the J-hook rotor through the previous analysis, two alternative rotor concepts were developed and evaluated using the same methodology. The following designs were assessed and compared against the J-hook baseline: a drilled rotor incorporating a pattern of through holes, and a slotted rotor featuring patterned through slots.

These designs were selected as they are widely adopted in motorsport applications for their ability to increase convective surface area, reduce rotor mass, and improve heat dissipation.

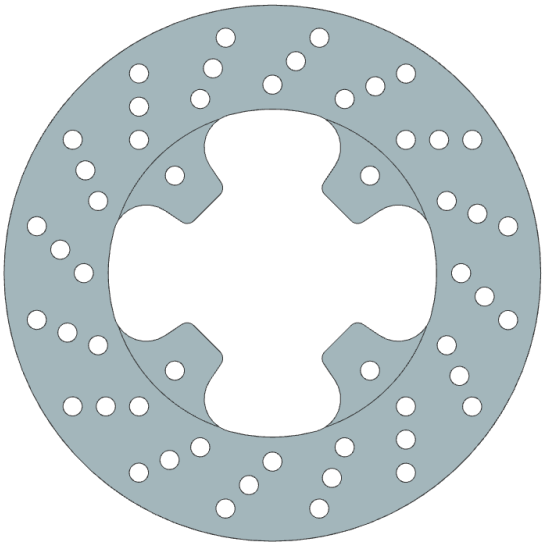


Figure 10: Drilled Brake Rotor Concept

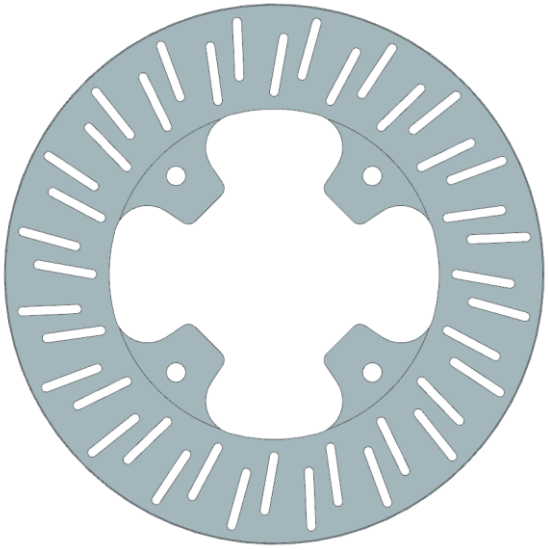


Figure 11: Slotted Brake Rotor Concept

Each alternative concept was subjected to the same methodology applied to the J-hook rotor. Transient thermal simulations were conducted on both designs, with the resulting temperature distributions imported into structural FEA and optimization simulations. This consistent approach allows for a comparison between all three rotor concepts against the defined design objectives.

4.6 Design Optimization and Selection

Once all simulation models were correctly running and providing results consistent with 2025 competition data, optimization could begin. The main goal of this project was to improve the braking performance of the rotors while keeping mass as low as possible. To achieve this, optimization in NX served as a useful tool for analysis. To lower rotor temperature, an increase in mass was required, but it was unclear which modification of brake rotor geometry would have the greatest effect on temperature. For example, increasing/decreasing thickness and diameter will both reduce temperature and mass at different rates. The purpose of completing optimization studies was to figure out which combination of these changes, paired with different cut patterns, would keep mass as low as possible.

To set these studies up, they were performed on the transient thermal studies listed above. As the structural FEA studies included thermal data by importing past thermal results, it was deemed not possible to incorporate the changing geometric effects on different temperature studies. Furthermore, structural FEA revealed that the brake and friction forces were negligible compared to the thermal studies.

Next, objectives, constraints, and design variables were selected, as shown in the figures below. The objective was set in NX to minimize weight. The constraint was then assigned as a model constraint, with the maximum temperature in the rotor body at any point in time not exceeding 500°C. Finally, design parameters were selected. NX was given the choice of varying both the

rotor thickness (labeled p35) between 4mm and 5.5mm, and the diameter (labeled p9) between175mm and 225mm.

Define Constraints

Name	Category	Response Type	Limit Type	Limit Value
Result Measure	Model Constraints	Result Measure	Upper	500.000000

Figure 12: Maximum Temperature Constraint

Define Design Variables

Name	Initial Value	Upper Limit	Lower Limit
"J-Hook Brake Rotor"...	5.101407	5.500000	4.000000
"J-Hook Brake Rotor"...	177.807171	225.000000	175.000000

Figure 13: Maximum Thickness and Diameter Variables

Next, the optimization was run. The first trial was run on a rotor with no hole pattern. This was to gain an understanding of the effect of each pattern's cooling effect as thickness and diameter were varied. The results can be found in the figures below.

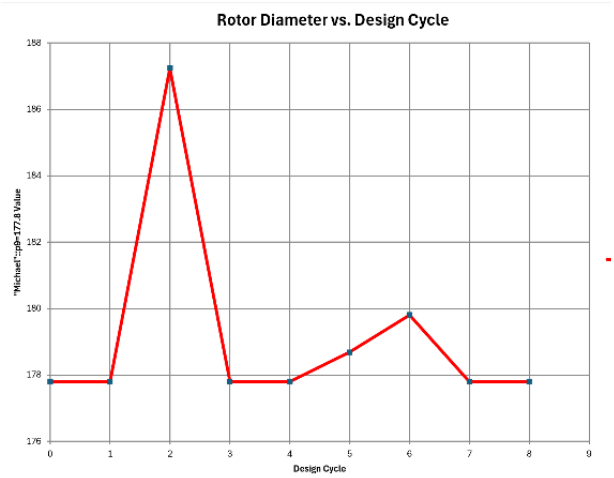


Figure 14: Blank Rotor Diameter Optimization

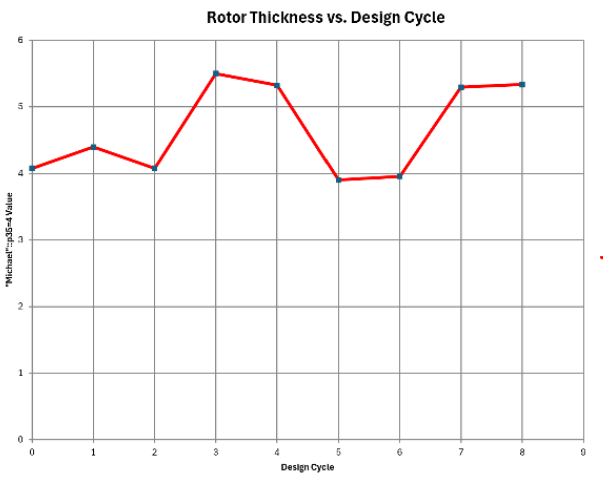


Figure 15: Blank Rotor Thickness Optimization

Table 4: Blank Rotor Optimization Results

Design Objective Function Results									
Minimum Weight [mN]	0	1	2	3	4	5	6	7	8
	5601.25	6040.96	6450.75	7558.22	7314.84	5433.8	5605.79	7275.41	7331.97
Design Variable Results									
Name	0	1	2	3	4	5	6	7	8
"Blank"::p35=4	4.07	4.39	4.07	5.49	5.32	3.90	3.95	5.29	5.33
"Blank"::p9=177.8	177.8	177.8	187.24	177.8	177.80	178	179.80	177.8	177.8

Design Constraint Results									
	0	1	2	3	4	5	6	7	8
Result Measure Upper Limit = 500 [°C]	615.21	593.54	625.65	569.64	565.63	622.0	621.56	569.95	565.24

The optimization study revealed that no combination of thickness and diameter would remain below 500 degrees. Thus, it was concluded that a hole pattern is necessary for optimal braking performance.

Next, optimization was performed on the J hook, drilled, and slotted style rotors. The constraints and variables remained the same when compared to the first optimization run. The results returned that a maximum temperature of 500 °C was possible with the hole patterns by increasing their thickness and diameters. Detailed optimization results for each rotor concept, including convergence graphs and output tables, are provided in the Appendix.

Table 5: Final Optimized Rotor Properties

	Blank	**J Hook**	**Drilled**	**Slotted**
Minimum Mass [kg]	0.799	0.642	0.660	0.497
Thickness [mm]	5.33	5.02	5.17	4.14
Diameter [mm]	177.80	175.00	177.8	177.81
Maximum Temperature [°C]	565.24	498.81	499.8	500.01
Maximum Stress [MPa]	1385.67	2384.47	1458.24	1475.43

The final optimization results point to the slotted rotor as the most effective design. It achieves the lowest mass while keeping temperature under control. Although the simulated results indicate that they perform the best overall, their cooling and braking performance should be validated experimentally before being used in a racing environment.

One key aspect of the slotted design is that it removes the greatest amount of pad contact area and overall rotor volume. This reduction in volume increases the surface area available for convective heat transfer in the simulation. For simplicity, the model assumed a uniform convection coefficient across all surfaces. While that assumption is reasonable, in practice, the convection coefficient will vary, which could either enhance or reduce the actual cooling benefits of these geometric features. Replicating the simulated braking events in a test and measuring the rotor temperatures would be an effective way to validate the model. Additionally, the reduced braking surface means that less pad and rotor contact area exists, which could impact braking performance. As this was not covered in these simulations, it would be important to note these findings with a test.

As a result, slotted rotors are selected as the best design but recommended for further testing, as both the transient thermal and structural results (which can be found in the Appendix) suggest they

offer the best performance with the greatest mass savings. Drilled rotors are a strong secondary option as they meet the temperature requirements without significantly reducing pad contact area, and do not show the same high stress concentrations that can lead to cracking over time.

5.0 Group Effort and Individual Contributions

The project was completed collaboratively between three group members, with responsibilities divided based on individual strengths while maintaining group-based decision making throughout the design and simulation process. The contribution of each member is outlined in the table below.

Table 6: Group Member Contributions Table

Task	Jackson Hamel	Michael Pemberton	Ben Stratford
CAD Modelling – Original Rotor			
CAD Modelling – Alternative Concepts			
Preliminary Calculations			
Transient Thermal Simulation			
Structural Simulation			
Optimization Study			
Report Writing			
Presentation Preparation			

While specific tasks were led by individual members, all major decisions were jointly agreed upon. All members contributed equally to the overall project outcome without any issues.

6.0 Summary

This projected conducted a thermal and structural analysis of an FSAE brake rotor using Siemens NX, motivated by operating temperatures exceeding 630°C during competition. Transient thermal simulations and structural FEA were performed on the existing J-hook rotor design to verify its thermal performance. Simulation results confirmed peak operating temperatures of approaching 600°C, with thermal loads identified as the dominant structural load. These findings validated the need for design improvement and established the methodology that was applied consistently across all rotor concepts.

Two alternative rotor concepts (drilled and slotted) were developed, simulated, and compared against design objectives of peak operating temperature, structural integrity, and minimal mass. Geometry optimization was completed to determine the minimum viable rotor mass. Based on the comparison of simulation results, the slotted rotor was identified as the optimal concept, achieving a peak operating temperature of 500°C and the most favourable mass and stress distribution of the three designs evaluated. The peak temperature and mounting compatibility objectives were fully met, with a mass reduction of 7.1% achieved relative to the original design. While this falls slightly short of the 10-15% target, further material removal was constrained by the need to maintain the peak operating temperature below 500°C. Reducing rotor mass beyond this point resulted in temperatures exceeding the thermal threshold. Further optimization of the slot pattern and rotor profile is recommended to achieve greater mass reduction without compromising thermal performance. The slotted rotor design is recommended for manufacturing and implementation on the UV26 FSAE vehicle for the 2026 competition season.

7.0 Experience and Suggestions

This project provided valuable experience with the complete CAD/CAE workflow within Siemens NX, from geometry modelling to thermal and structural simulation. Working through the simulation process developed a strong understanding of how CAD/CAE tools are used in real engineering work.

This project also highlighted the importance of correctly defining boundary conditions, as small errors or inconsistencies in heat flux application, convection setup, or constraint definition, had a large impact on simulation results. Validating the simulation results against real competition data provided a meaningful metric and reinforced confidence in the methodology utilized.

Several areas of further development were identified beyond the scope of this project. As this analysis was conducted within a CAD course context, manufacturing considerations were not fully explored. Future work could include optimization studies solved for nominal sheet steel stock sizes commercially available to reduce manufacturing complexity. From a manufacturing standpoint, the rotor profile is well suited to waterjet cutting, requiring only a single DXF file and reference drawing for production, making manufacturing straightforward and less costly. Lastly, further geometric optimization could be performed on the rotor pattern profile to identify the optimal hole geometry, slot geometry, or combination of both that improves thermal performance and mass reduction.

Overall, the group achieved its main objective. Through CAD/CAE simulation, thermal performance of the existing rotor was validated and addressed through the development and evaluation of improved design concepts that reduced peak operating temperatures and better meet the demands of FSAE competition.

8.0 References

[1] "Brembo Genuine Sintered 07BB209A Rear Brake Pads," Moto24.EE, [Online]. Available: https://moto24.ee/brembo-genuine-sintered-07bb209a-rear-brake-pads.

[2] M. Otkur, I. Fasahi, T. Waseem, O. Zattam and M. Alazmi, "Disc Brake Rotor Thermal Analysis for a Formula SAE Race Car," Proceedings of the 10 th World Congress on Mechanical, Chemical, and Material Engineering (MCM'24), Barcelona, 2024.

[3] P. Baskara Sethupathi, A. Muthuvel, N. Prakash and L. Stanly Wilson, "Numerical Analysis of a Rotor Disc for Optimization of the Disc Materials," Journal of Mechanical Engineering and Automation, 2015.

9.0 Appendix

9.1 J Hook Rotor Optimization Data

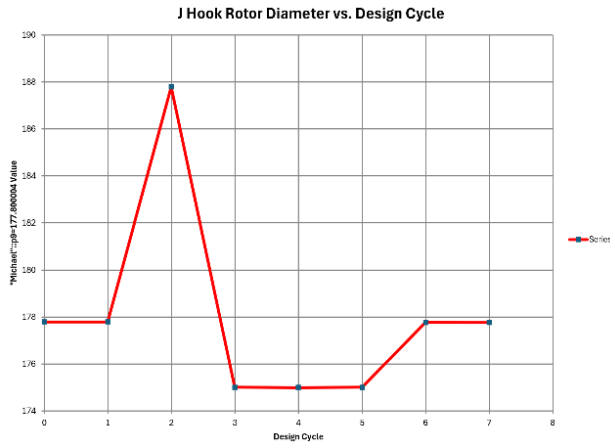


Figure 16: J Hook Rotor Diameter Optimization

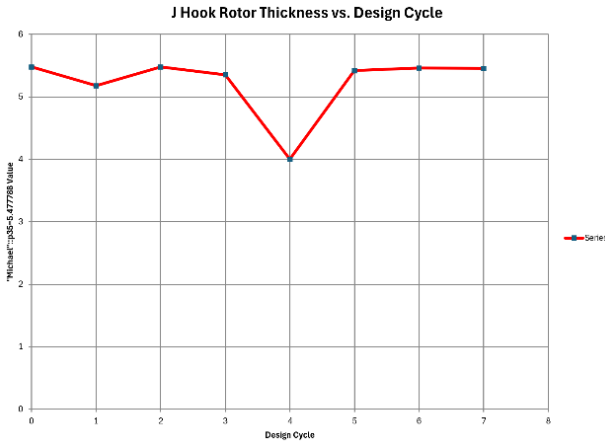


Figure 17: J Hook Rotor Thickness Optimization

Table 7: J-Hook Rotor Optimization Results

Design Objective Function Results								
Minimum Weight [mN]	0	1	2	3	4	5	6	7
	7199.1	6805.6	8410.5	6716.8	5018.8	6799.4	7171.8	7165.9

Design Variable Results								
Name	0	1	2	3	4	5	6	7
"J-Hook"::p35=5.47	5.48	5.17	5.47	5.3526	4.0000	5.4187	5.4594	5.45516
"J-Hook"::p9=177.80	177.8	177.8	187.8	175.01	175.0013	175.0107	177.77	177.76
Design Constraint Results								
	0	1	2	3	4	5	6	7
Result Measure Upper Limit = 500.000000 [°C]	499.21	516.01	521.58	503.77	539.8	505.69	499.57	499.73

9.2 Drilled Rotor Optimization Data

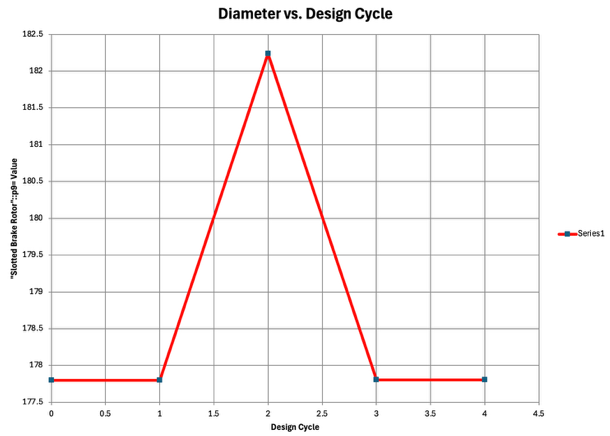


Figure 18: Drilled Rotor Thickness Optimization

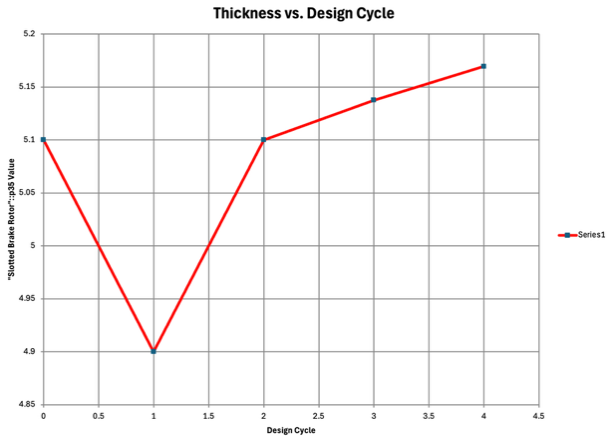


Figure 19: Drilled Rotor Diameter Optimization

Table 8: Drilled Rotor Optimization Results

Design Objective Function Results					
Minimum Weight [mN]	0	1	2	3	4
	6386.329	6135.896	6879.424	6433.769	6473.737
Design Variable Results					
Name	0	1	2	3	4
"Optimized Brake Rotor"::p35=5.1	5.1	4.9	5.1	5.137527	5.169438
"Optimized Brake Rotor"::p9=177.8	177.8	177.8	182.24	177.8041	177.8041
Design Constraint Results					
	0	1	2	3	4
Result Measure					

Upper Limit = 500.000000 [°C]	502.31	515.02	528.05	500.96	499.81

9.3 Slotted Rotor Optimization Data

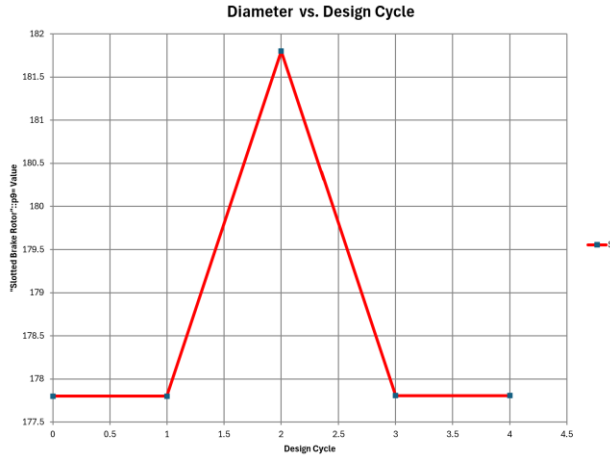


Figure 20: Slotted Rotor Thickness Optimization

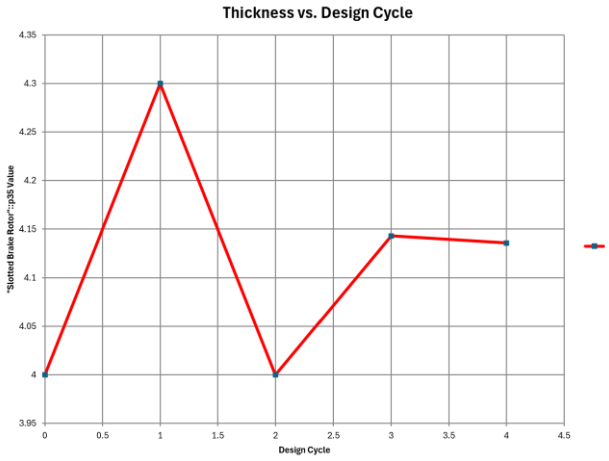


Figure 21: Slotted Rotor Diameter Optimization

Table 9: Slotted Rotor Optimization Results

Design Objective Function Results					
Minimum Weight [mN]	0	1	2	3	4
	4717.68	5071.587	5072.811	4888.682	4879.732
Design Variable Results					
Name	0	1	2	3	4
"Slotted Brake Rotor"::p35=4	4	4.3	4	4.1432	4.135871
"Slotted Brake Rotor"::p9=177.8	177.8	177.8	181.8	177.8081	177.8072
Design Constraint Results					
	0	1	2	3	4
Result Measure					
Upper Limit = 500.000000 [°C]	506.5	492.89	479.84	499.64	500.01

9.4 Optimized Rotor Thermal Distribution

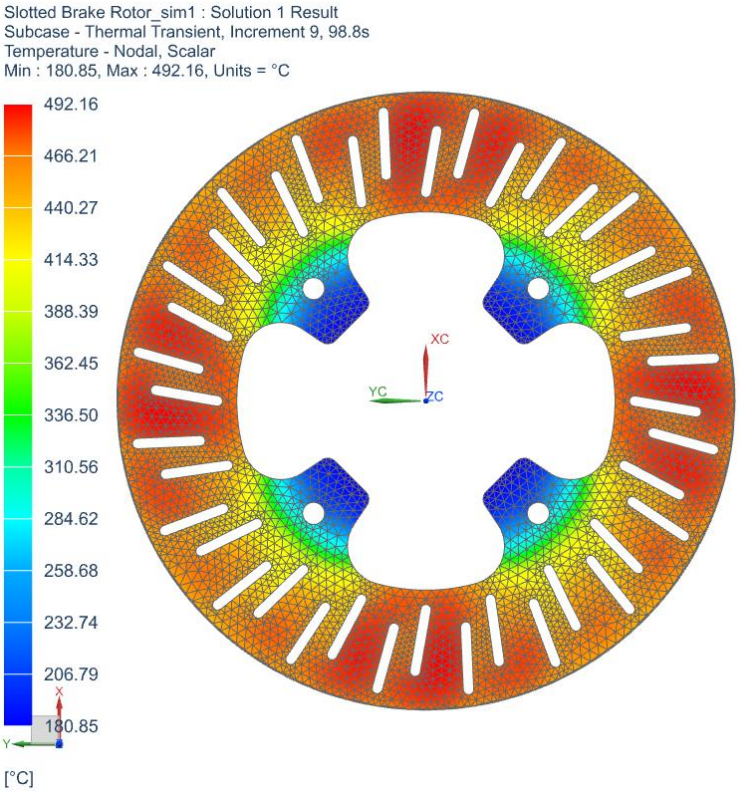


Figure 22: Optimized Rotor Thermal Distribution

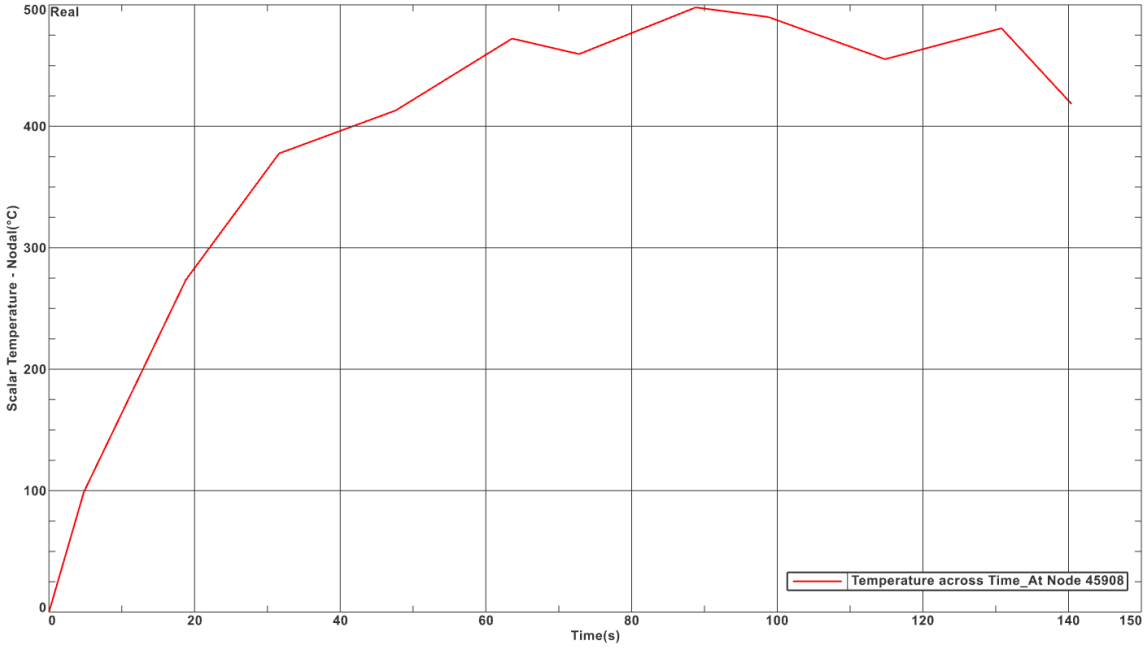


Figure 23: Optimized Rotor Temperature vs. Time Plot

9.5 Optimized Rotor Stress Distribution

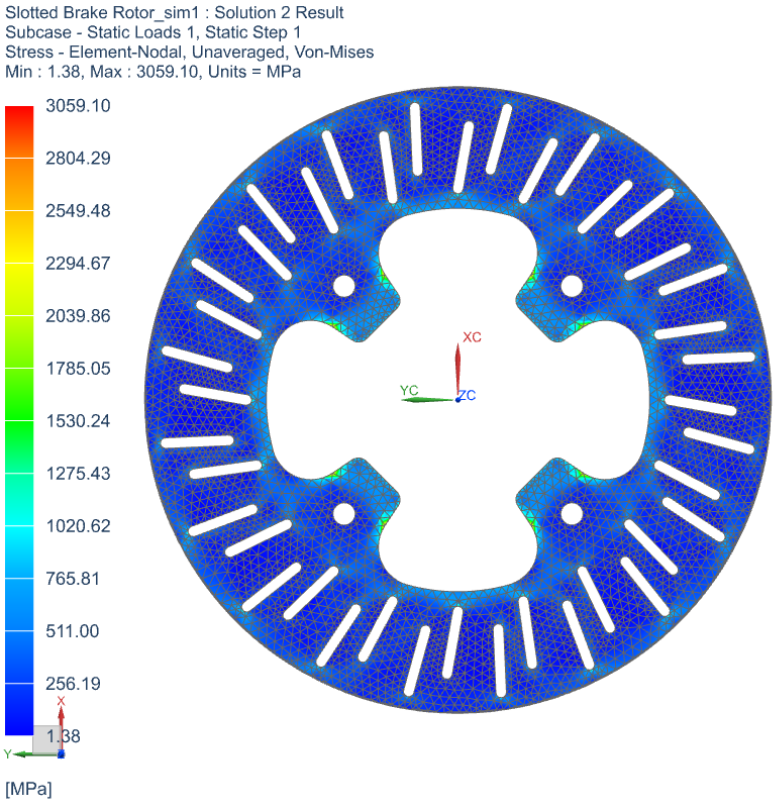


Figure 24: Optimized Rotor Stress Distribution